Financial Engineering Project

Making Concentrated Liquidity Pay

Risk-Sensitive Reinforcement Learning on Uniswap v3

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Abstract

Providing liquidity on Uniswap v3 looks passive, but it is not. A liquidity provider (LP) chooses a price range, earns fees while the price stays inside it, and loses whenever informed traders move the pool price against the quoted level. Those losses are not hypothetical: on many large pools the arbitrage losses borne by liquidity providers exceed the fees they earn. The LP is left with three decisions, all of them costly: when to recenter, how wide to keep the band, and how much of the position's exposure to hedge. We study those decisions as a risk-sensitive reinforcement learning problem, in a testbed built on three years of recorded on-chain trades, with the final year held out. An exact replay engine provides the ground truth, a fast simulator provides the training throughput, and a parity harness keeps the two honest. The report asks four questions. Where does fee income fall short of the losses? Why does a standard learning algorithm fail on this task, and what repairs it? Can a single policy manage timing, width and hedging together and outperform the best rules written by hand on unseen data? And under which market environment does active management begin to pay? [PENDING: The final numbers and the last two answers will be filled in once the experiment runs complete.]

Index Terms: reinforcement learning, Uniswap v3, concentrated liquidity, market making, loss-versus-rebalancing, impermanent loss, risk-sensitive reinforcement learning.

I Introduction

Automated market makers let anyone earn trading fees without quoting prices by hand. Uniswap v3 sharpened the offer: a liquidity provider (LP) chooses the price range it is willing to trade in, so a deposit of 100 000 USDC earns the fee share of a larger full-range position while the price stays inside the chosen band.[1] Narrowing the band raises the fee earned per dollar of capital, and that promise drew liquidity into v3.

The promise hides a cost. Liquidity quoted away from the market price is an offer that arbitrageurs accept, and every such trade pushes the pool toward the market price while leaving the LP holding more of the asset that is falling. Because the loss is a transfer rather than a fluctuation, it does not reverse when the price returns. Milionis et al. named this adverse selection cost loss-versus-rebalancing (LVR) and showed that a full-range position loses value at a rate proportional to the square of volatility.[2]

Fritsch and Canidio measured realised arbitrage losses on large pools and found that they exceed the fees those pools earn,[3] and Loesch et al. showed that concentration magnifies the loss once the price leaves the band.[4] Fee income alone is not the LP's return.

An LP who wants more than a passive range has three decisions, and they interact. When to recenter trades fee income against gas and slippage; how wide to set the next band trades fee density against the time until an edge; how much of the position to hedge trades directional risk against funding and perp fees. Each decision is an impulse, costly and taken under uncertainty about volatility, order flow and regime. The objective is not to earn the most but to earn the most per unit of risk taken, which places the problem within risk-sensitive impulse control rather than static portfolio choice.[5]

Recent work takes up parts of it. Zhang et al. train a hedged agent for v3 provision, but the hedge executes without cost or funding and the policy sees no regime information.[6] ZeroSwap learns to set a market maker's spread so that quoted prices track an external reference, a different lever from range placement.[7] RAmmStein derives regime-dependent rebalancing thresholds and reports fewer rebalances for the same net return, but its net metric excludes impermanent loss and it works with off-chain data for a single pair.[5] Chionas et al. study risk aversion in concentrated liquidity inside a simulation and leave out on-chain data and the hedge.[8] The closest study learns when to redeploy and how wide to stay on the same primary pool used here, but it leaves the hedge out and reports cumulative reward only.[9] What none of these provides is the combination this report is designed to test: one risk-sensitive policy over timing, width and hedge, every friction charged on both legs, evaluated against hand-written rules on the trades that actually occurred.

This report fills that gap. We build a testbed on three years of recorded trades from two real ETH/USDC pools, with an exact replay backtester that computes fees in the protocol's own integer arithmetic, a fast simulator for training, and a parity harness that keeps the two honest on identical inputs. On that testbed we ask four questions.

- **Where do fees fail?** We decompose passive returns into fees, impermanent loss, gas and slippage, and locate the conditions under which the losses win.
- **Why does naive learning fail?** We document the collapse of a standard policy on this task, trace it to three interface problems, and measure what each repair buys.
- **Can one policy do better?** We train a risk-sensitive policy over timing, width and hedging and compare it out of sample with the best hand-written rules and a hedged benchmark.
- **Where is active management worth it?** We map the boundary between active and passive provision across volatility, fee tier, gas and capital.

The failures of the first policies are part of the result, not a footnote. In this domain, what the agent is allowed to see and what it is rewarded for matter more than the choice of learning algorithm, and a report that showed only the eventual policy would hide the more transferable lesson.

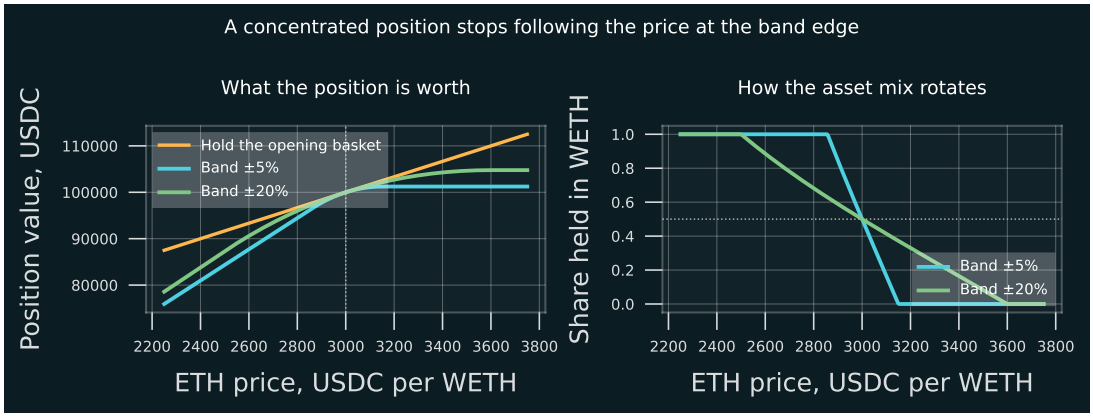


Figure 1: A concentrated position versus a passive holding of the same opening basket. Left: the position tracks the basket while the price sits inside the band and stops following it at the edges, so it gives up the upside of a rally while keeping the downside of a fall. Right: the asset mix rotates from WETH to USDC as the price rises and freezes at the edge. Illustrative parameters: 100 000 USDC entered at 3000 USDC per WETH.

II How Concentrated Liquidity Works

II.A A position as a bet on a range

Uniswap v3 indexes price by integer ticks. An LP picks a lower and an upper tick, and the deposit backs trades only while the pool price stays between them.[1] In protocol units the price is a raw square root, $s = \sqrt{1.0001^{\text{tick}}}$, which increases with the tick; the human price is $q(s) = 10^{d_1-d_0}/s^2$ USDC per WETH, so a lower tick means a higher ETH price. Inside a band $[s_a, s_b]$ the position holds the token amounts

$$x(s) = \frac{L}{10^{d_0}}\left(\frac{1}{s} - \frac{1}{s_b}\right), \tag{1}$$

$$y(s) = \frac{L}{10^{d_1}}\left(s - s_a\right), \tag{2}$$

and its USDC value is therefore

$$V(s) = x(s) + y(s)\,q(s) = \frac{L}{10^{d_0}}\left(\frac{2}{s} - \frac{1}{s_b} - \frac{s_a}{s^2}\right). \tag{3}$$

As the ETH price falls, the position sells USDC and accumulates WETH; as the price rises it does the opposite. At an edge of the band the position holds a single asset, and once the price leaves the band it stops trading altogether, so a passive position cannot escape the wrong side of a move.

Fees accrue in proportion to the position's share of active liquidity. Narrowing the band raises that share for a given deposit, which is why concentration is attractive, but it also shortens the expected time until the price reaches an edge. That tension between fee density and band lifetime sits underneath every decision in the report, and Fig. 1 shows the two effects together.

Table 1: The three levers and the trade-off each carries. Every return in the study is measured after the costs of the active levers are charged: the liquidity leg pays gas, slippage and impermanent loss, and the hedge leg adds funding and perp fees once the overlay is enabled.

Lever	Choice	Buys	Costs
Timing κ_t	hold or recenter	fee income on a band close to the price	gas and slippage; the fee position restarts
Width w_t	width of the band, in tick spacings	fee density for a given deposit	time until the price reaches an edge
Hedge h_t	short fraction of ETH delta	removal of directional swings	funding and perp trading fees

II.B Two losses, one larger

Impermanent loss (IL) is the shortfall of the position against holding the initial deposit. For a full-range position with equally valued legs and price ratio $r = q_t/q_0$,

$$\mathrm{IL}(r) = \frac{2\sqrt{r}}{1+r} - 1, \tag{4}$$

which is second order in the price move; at $r = 1.2$ it is -0.41 %. On v3 the same expression applies only inside the band, and outside it the position is a one-sided holding, so IL depends on where the band was placed.[4]

LVR is sharper. Compare the position with a portfolio that continuously rebalances to hold the same instantaneous exposure. A second-order expansion of the full-range value shows that the LP falls behind that portfolio at the rate

$$\mathrm{dLVR}_t = \frac{1}{8}\,\sigma^2\,V_t\,\mathrm{d}t, \tag{5}$$

the loss-versus-rebalancing of Milionis et al.[2] Adverse selection therefore grows with the square of volatility and with position size, and fee income must exceed it for passive provision to be viable. Concentration raises fee density but not the fee rate, so it helps only when the extra fee share outweighs the extra exposure to arbitrage; Section III finds that on many large pools it does not. Figure 2 shows the shape of both losses.

II.C The decisions and the risk

An LP that wants to improve on the passive outcome must decide when to act, where and how wide to place the next band, and how much market risk to carry. The levers interfere, and Table 1 states what each one buys and what it costs. Because acting is lumpy, the problem is one of impulse control,[5] and because the outcome is uncertain the right measure is profit per unit of risk. We use the annualised Sharpe ratio and the conditional value-at-risk (CVaR) at the 5 % level as primary statistics; CVaR is coherent and has a convex representation that can be optimised directly.[10] Figure 3 makes the sequence of decisions and costs concrete.

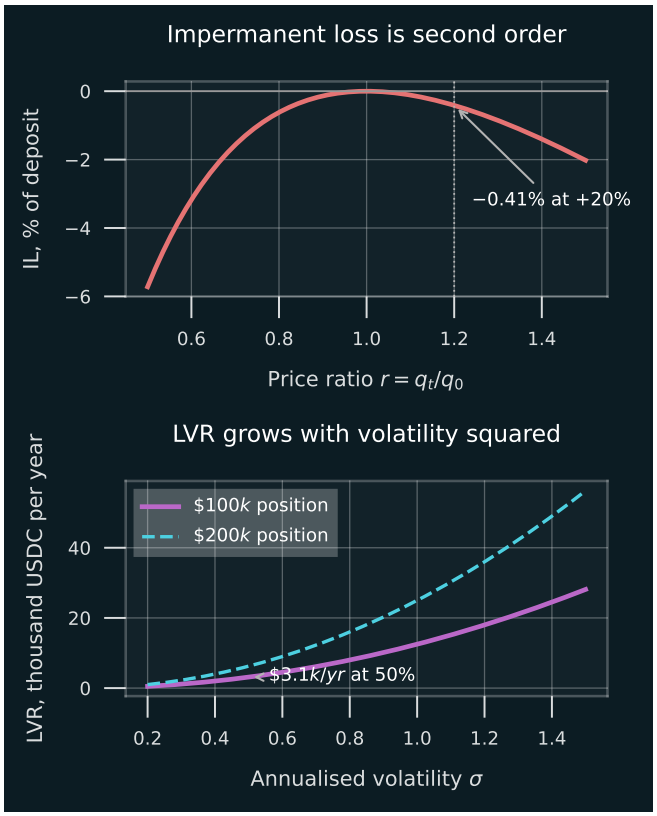


Figure 2: The two losses. Top: impermanent loss against the price move, second order and small for a full-range position. Bottom: the LVR rate against volatility for a 100 000 USDC and a 200 000 USDC position, showing both the quadratic dependence on volatility and the linear dependence on size.

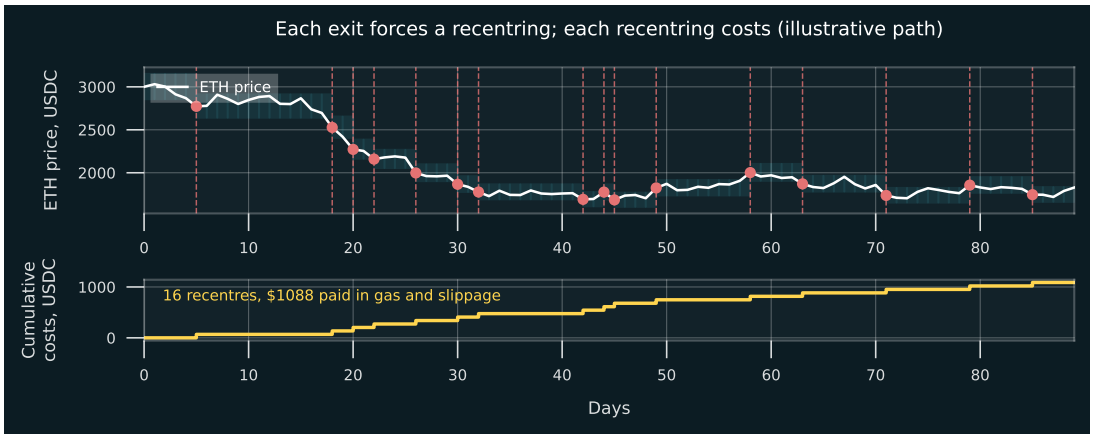


Figure 3: Why acting is a decision rather than a rule. An illustrative price path with a ±5% band: each exit forces a recentring, and each recentring adds gas and slippage to a cost that only ever grows. The path is illustrative and the costs are simplified; the study charges per-block gas and slippage on the inventory actually swapped.

III Related Work

The economics of passive provision is now well understood. Milionis et al. formalised the loss a constant-product LP suffers against a delta-rebalanced portfolio and derived the rate $\frac{1}{8}\sigma^2 V$ for a full-range position; the loss appears on any price path, informed traders or not, and protocol fees and the choice of curve shift only the balance.[2] Fritsch and Canidio measured the losses on real pools and found that arbitrage losses typically dominate the fees earned, with passive full-range positions outperforming concentrated ones in their sample.[3] Loesch et al. studied v3 positions directly and showed that concentration magnifies impermanent loss when the price leaves the band.[4] Together these results define the terrain: gross fee yield is not an LP's return.

A second line of work learns the LP's decisions. Zhang et al. train a hedged double deep Q-network and report large gains over baselines, but the hedge is assumed frictionless and the policy sees no regime features.[6] A recent replication replaces the feature extractor of that model with a sequence model and a revised reward, and adds two baselines; it keeps the simulated setting and reports no statistic that adjusts for risk.[11] ZeroSwap learns the spread of a market maker so that quoted prices track an external reference, an orthogonal lever to range placement.[7] RAmmStein poses regime-dependent rebalancing thresholds as a problem of impulse control and reports that rebalancing lazily when the regime warrants it improves net ROI with far fewer rebalances, though its net metric excludes impermanent loss and the study relies on off-chain data for one pair.[5] Chionas et al. treat concentrated liquidity as a risk-aversion problem in simulation and find that higher volatility and gas widen optimal positions and reduce rebalancing, while a risk penalty trims the adverse tail; their analysis uses no on-chain data and no hedge.[8] Classical market making supplies the risk-sensitive vocabulary we use, but its state is a limit order book and its actions are quotes, not range geometry.[12]

The closest study to this report trains a PPO agent on hourly WETH/USDC 0.05 % data from the same primary pool used here.[9] Xu and Brini let the policy decide when to redeploy and how wide to keep the band, penalise the loss-versus-rebalancing rate inside the reward, and compare the learned policy with a periodic reset across eleven rolling windows; the active policy wins seven of them. Three differences matter for the claims of this report. Their agent carries no hedge, so directional risk stays in the position. Market outcomes are simulated from closed-form fee and loss expressions at hourly resolution, with a fixed gas charge and no slippage, rather than replayed from the recorded events. And the evaluation is thinner: performance is reported as cumulative reward, without a statistic that adjusts for risk or an uncertainty interval, and the best of fifty trained agents is selected for each window.

The learning machinery is standard. We use proximal policy optimisation with clipping (PPO),[13] implemented through Stable-Baselines3 on Gymnasium environments,[14,15] and evaluate risk with CVaR at the 5 % level through its Rockafellar–Uryasev representation.[10]

No study surveyed here combines the parts this report tests. The line that learns decisions supplies them only in pieces: a hedge learned without friction, a range policy learned without the hedge, and a mean metric rather than one that adjusts for risk. Two of the answers here are measured rather than assumed. The passive benchmarks are decomposed into fees, impermanent loss, gas and slippage on both fee

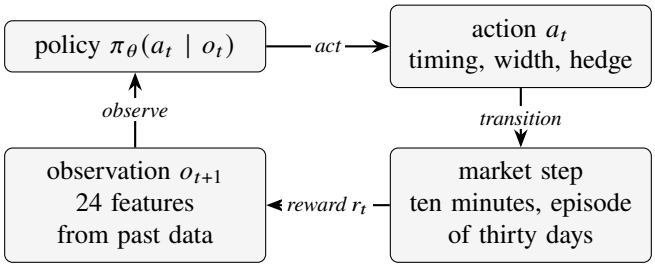


Figure 4: One cycle of the LP decision process. The policy acts every ten minutes; a decision to rebalance moves the band and charges gas and slippage, while a decision to hold leaves the position where it is. Rewards accumulate over an episode of thirty days against the fixed opening basket.

tiers, so the economics answer rests on the pools of this study rather than on a survey, and the ways a standard policy collapses on the task are reported as a result rather than hidden in an implementation note.

What remains is the combination studied here. One risk-sensitive policy chooses timing, width and hedge. A cost ledger replayed from the tape, bounded against an exact integer engine, prices every friction on both legs. Sharpe and tail statistics carry paired intervals across seeds. And a boundary states where active management stops paying. The rest of the report tests that combination, with the hedge leg the one part still pending: the overlay is disabled in the shipped configuration, so the hedge results are marked as such.

IV The Liquidity Provider's Decision Problem

We model the LP's problem as a risk-sensitive Markov decision process with impulse control. Decisions are taken every ten minutes over an episode of thirty days. At each decision the agent sees a fixed observation, chooses between holding and acting, and receives a reward that measures how far the strategy's value has moved beyond a passive benchmark, net of every cost. The volatility regime and order flow are latent, so the process is partially observed: the agent sees only the history summarised in the observation. Figure 4 shows one decision cycle.

IV.A State

The observation contains 24 features, all computed from information available at the decision time. Table 2 groups them by source. The band geometry deserves emphasis: the action space is written in tick spacings relative to the current tick, so a policy that cannot see how close the price is to an edge cannot time a rebalance at all. Heavy-tailed features such as price, liquidity and gas are compressed with a signed logarithm so that every dimension reaches the network on the same scale.

Table 2: The 24 observation features, grouped by source. Every field is computed from information available at the decision time, so an observation at time t cannot contain a bar from $t + 1$.

Group	Features	n
Market	price, sqrt price, tick, 1 h and 24 h log returns, realised 24 h volatility	6
Position	in-range flag, position value, two uncollected fee balances, IL against the fixed basket	5
Pool	active liquidity, fee tier	2
Costs	recent gas price, recent ETH reference price	2
Regime	regime label (bull, bear, sideways, high volatility)	1
Time	step in episode, steps remaining	2
Band	distance to each band edge and band width, in tick spacings	3
Hedge	short ratio, hedge equity, recent funding rate	3
	Total	**24**

Table 3: The four action heads. Widths are given as half the band width, in tick spacings. The primary pool's spacing of 10 uses all ten widths; the contrast pool's spacing of 60 uses the six-width ladder.

Head	Symbol	Options	Meaning
Rebalance	κ_t	hold, act	spend gas and slippage to move the band, or leave it
Centre	c_t	$-4, \ldots, 4$ spacings	shift the centre of the new band around the current tick
Width	w_t	$\{1, 2, 5, 10, 25, 50,$ $100, 250, 500, 1000\}$ spacings	half of the new band's width
Hedge	h_t	$\{0, 0.25, 0.5, 0.75, 1.0\}$	fraction of the LP's ETH delta carried by the short

IV.B Action

At each decision the agent selects a factored action

$$a_t = (\kappa_t,\, c_t,\, w_t,\, h_t), \tag{6}$$

whose four heads are described in Table 3. On the 0.05 % primary pool the liquidity head alone has $1 + 9 \times 10 = 91$ options. The shipped configuration learns the liquidity head with the hedge overlay disabled; when the overlay is enabled its ratio is set by configuration. Promoting h_t to a learned head, and the hedge-leg benchmarks, are planned for the last two questions and their results remain pending.

IV.C Reward

A rebalance converts inventory and pays for the privilege. Writing V_t for the position value, H_t for the fixed value of the opening deposit held passively, and the hedge cash flows as mark PnL π_t, funding ϕ_t and perp cost $\tau_t \le 0$, the reward is

$$r_t = \frac{1}{W_0}\Big(\text{fees}_t - \text{gas}_t - \text{slip}_t + \Delta(V_t - H_t) + \pi_t + \phi_t + \tau_t - \lambda\,\hat{\sigma}_t\Big), \tag{7}$$

Table 4: Every term of the reward, in token0 (USDC) units before the $1/W_0$ scaling. Gas and slippage enter as positive magnitudes and carry a minus sign in the reward.

Term	Meaning	Sign
fees_t	fee income earned in the step	+
gas_t	gas for mints, burns, collects and rebalancing swaps	−
slip_t	slippage on the inventory actually swapped	−
$\Delta(V_t - H_t)$	change in position value against the fixed opening basket	±
π_t	mark PnL of the short perpetual	±
ϕ_t	funding paid or received on the short	±
τ_t	perp taker cost of adjusting the short	≤ 0
$\lambda\,\hat{\sigma}_t$	penalty on recent reward volatility, $\lambda = 0$ by default	−

whose terms are defined in Table 4. Dividing by the initial capital W_0 keeps value targets at a workable scale without hiding any cost. The excess is measured against the *fixed* opening basket: H_t is never rebased at a rebalance, so the episode sum equals the excess over holding that basket, which is the quantity the evaluation reports.

IV.D Risk and benchmarks

The learning objective is the discounted return with a penalty on recent reward volatility: the risk term in Eq. (7) subtracts a multiple of the trailing standard deviation of per-step equity changes, with $\lambda = 0$ by default. Tail risk is measured in the evaluation instead. With losses $L = -G$ defined on the episode return G, the conditional value-at-risk at level α is

$$\text{CVaR}_\alpha(L) = \min_{\nu \in \mathbb{R}} \left\{ \nu + \frac{1}{1-\alpha} \mathbb{E}\left[(L - \nu)^+ \right] \right\}, \tag{8}$$

the Rockafellar–Uryasev representation,[10] which we report at $\alpha = 0.95$. Evaluation therefore reports the annualised Sharpe ratio and CVaR at 5 % as primary statistics, with raw PnL and excess over HODL as secondary ones. Comparisons use paired per-window differences and confidence intervals across seeds.

The hedge is a short perpetual position sized as a fraction of the LP's delta. Its mark PnL over a step is the negative product of the short notional and the price change, plus funding and the taker cost of adjusting the position. Three reference points make the hedge leg interpretable: the unhedged LP, the deterministic overlay, and a hedged HODL portfolio with matched beta. The last is the benchmark that prevents us from declaring success against a weaker opponent: a policy that merely removes directional risk must still beat a passive portfolio with the same risk profile. The overlay is disabled in the shipped configuration, so the hedged benchmark of Table 7 remains pending.

V Building a Trustworthy Testbed

A study of this question needs a world in which the LP's costs can be measured rather than assumed. That world must be exact enough that a reported fee or loss is the real one, and fast enough that a policy

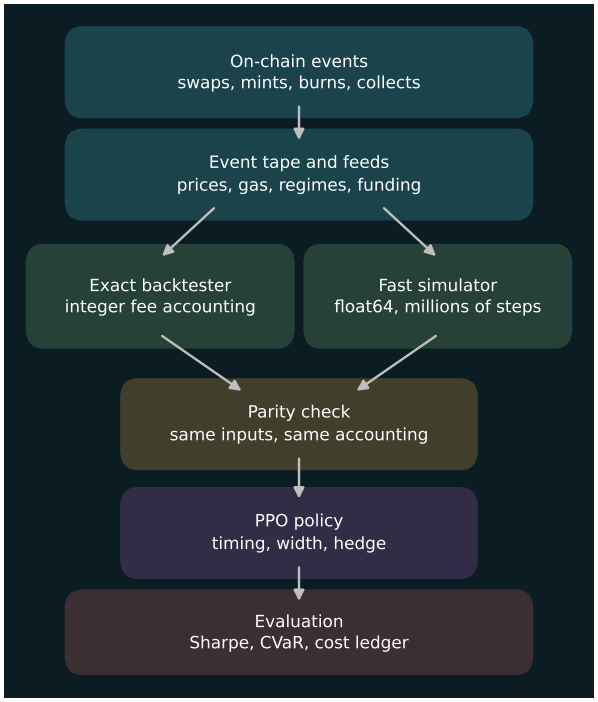


Figure 5: The testbed. Recorded events feed one tape; two engines consume it, an exact backtester for ground truth and a fast simulator for training; the parity check reconciles them before a policy is trusted.

can make millions of decisions. No single engine satisfies both, so the testbed uses two and checks them against each other. Figure 5 shows the flow.

V.A The record

The data are the raw logs of two real ETH/USDC pools: the 0.05 % pool used as the primary market and the 0.30 % pool used as a fee-tier contrast. The pipeline turns those logs into one ordered event tape of swaps, mints, burns and collects, and joins it with three auxiliary feeds: one-minute reference prices, per-block gas prices, and regime labels computed from thresholds fixed in advance. A fourth feed, the funding rate of a perpetual margined in USDT, prices the hedge. The window runs from January 2022 to December 2024, with 2022–2023 for training and 2024 held out. The primary pool's training tape alone holds about 4.86 million events, which is why speed matters later. Every read goes through one interface that enforces block ordering and the train/eval boundary, so an observation at time t cannot accidentally see a bar from $t + 1$.

V.B The exact engine

The backtester replays a frozen policy against the tape in the protocol's own integer arithmetic. Fees come from the pool's recorded fee-growth accumulators, exactly as the contract computes them; liquidity events update an external tick lattice seeded from a boundary snapshot; swaps are apportioned across the ticks they cross. The ledger therefore satisfies an accounting identity by construction: fees plus impermanent loss, minus gas and slippage, equals the strategy's excess over holding the opening

deposit. A backtested policy is frozen by construction because the interface exposes no way to update it, and every ledger records the configuration and seed that produced it.

V.C Two engines, one check

Training on the exact engine would be far too slow, so the simulator mirrors the same economics in floating point. Its agent is marginal: the position never moves the price, and its fee share is the fraction of active liquidity it supplies. The simulator can run on the recorded events or on a calibrated regime-switching jump process fitted to the training years, and it executes each action at the price observed before the interval advances, as a live LP would.

Two numeric worlds are only useful if their disagreement is bounded. The parity harness feeds both engines the same prices, positions and gas series, and then compares fees, impermanent loss and PnL under a stated tolerance. Since the simulator's fee engine was made exact, the remaining drift on real data is tiny: mean fee differences on the order of 10^{-7} USDC per step per seed, and impermanent loss matching exactly. The harness originally exercised only the passive policy, which never rebalances; it now also runs a policy that does, which is the path the fee and loss results depend on. A separate suite of look-ahead probes attacks the composed system at the train/eval seam.

Other testbeds make different trade-offs. SAiFE-gym exposes the market as a model: price impact, liquidity depth and gas enter as components, and the environments are vectorised for throughput.[16] A second line reconstructs the pool's liquidity state parametrically from swap activity and reports average fee approximation errors of about two percent on historical pools.[17] The first buys coverage where recorded state is missing, since a model can price liquidity that never existed on chain; the second recovers an approximate state from the trades themselves. The two pools of this study have their recorded state, so the backtester replays that state and the floating-point engine is used only where throughput is needed, with the parity check bounding the substitution.

V.D The failures that shaped the design

The first two training runs produced policies that did nothing useful: every seed collapsed to one deterministic action, some always holding and some always repeating a single rebalance, and the measured advantage barely distinguished the actions. The collapse was not a tuning problem. Table 5 lists the interface defects behind it and the repair for each. Each repair is small, but together they were the difference between a policy that can learn and one that cannot, which is why the failure is treated as a result rather than an implementation note.

VI Experimental Design

The protocol is fixed before any result is inspected. Table 6 states the pools, the split and the training budget, and Fig. 6 shows how the windows sit on the calendar.

Table 5: The defects that collapsed the first policies, and the repair for each. The first three are interface problems; the last two surfaced while fixing them.

Defect	Consequence	Repair
Observation had no band geometry	the policy could not tell how close the price was to an edge or how far a recentring would move, so it could not time a rebalance	distances to each edge and the band width, in tick spacings
Reward measured loss against a rolling entry basket	after a rebalance the reward no longer matched the reported metric	excess against the fixed opening basket, never rebased
Every episode replayed one fixed window	the agent never saw a different market	training windows resampled from 2022–2023
Slippage charged on the whole position value	costs overstated against a real swap	slippage on the inventory actually swapped
Fee credit taken from global fee growth	a position could be credited more than the fee actually taken when its liquidity dominated a thin segment	the position's exact share of each segment's LP fee

Table 6: The study protocol and the training configuration. Both pools are USDC/WETH with USDC as token0; prices are quoted in USDC per WETH. Contract addresses are Ethereum mainnet and are abbreviated.

Setting	Value
Primary pool	0.05 % (500 pips), tick spacing 10, `0x88e6a0…f5640`
Contrast pool	0.30 % (3000 pips), tick spacing 60, `0x8ad599…e6d8`
Training window	2022-01-01 to 2023-12-31 UTC
Evaluation window	2024-01-01 to 2024-12-31 UTC
Episode	thirty days, one decision every ten minutes
Capital	100 000 USDC per episode
Evaluation tiling	12 windows of 30 days plus a partial final window (13 episodes)
Seeds	5
Training budget	1 000 000 timesteps per seed
Algorithm	clipped PPO, $\gamma = 0.99$, GAE $\lambda = 0.95$, clip 0.15
Network	2 hidden layers of 256 units
Parallel environments	4

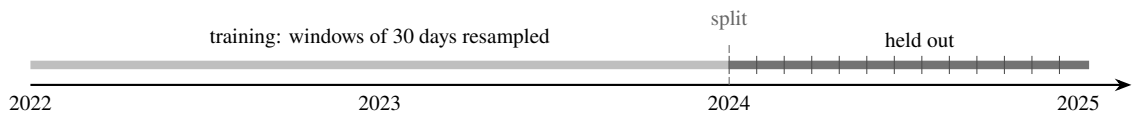


Figure 6: The walk-forward split. Training episodes are windows of thirty days sampled from 2022–2023; evaluation tiles 2024 into windows of thirty days, the last a partial period, and no episode crosses the split.

Table 7: Benchmark policies. The learned policy is compared with all of them; a positive claim requires beating the best static width and the hedged HODL with matched beta, not just hodl. The hedged HODL comparator is part of the planned protocol and runs once the hedge overlay ships.

Benchmark	Rule
hodl	Hold the initial token mix; never provide liquidity.
passive_narrow	Deploy a ±5 % band once; never recenter.
passive_wide	Deploy a ±20 % band once; never recenter.
full_range_v2	The Uniswap-v2-equivalent full range.
tau_reset	Recenter whenever the price exits the live band.
cost_aware	Recenter at most daily, and only when the estimated fee gain exceeds gas.
Best static width	The ladder width with the best Sharpe on the training split, frozen for test.
Hedged HODL	Initial basket plus a short perp matched to its beta, with funding and perp costs.
Learned policy	PPO over the joint (timing, width, hedge ratio) action space.

VI.A Benchmarks

A learned policy that beats only the weakest rule written by hand has demonstrated nothing. The comparison therefore includes the full ladder in Table 7, a best static width chosen on the training split and then frozen, and a hedged HODL portfolio with matched beta, and the learned policy must beat the strongest of these, not merely a passive holding. The hedged HODL benchmark runs once the hedge overlay ships; its values are pending.

VI.B Metrics

Two primary statistics carry the inference: the annualised Sharpe ratio, computed on each episode's equity curve and paired across seeds and windows, and CVaR at the 5 % level, the average of the worst 5 % of episode returns pooled across seeds. Both are reported with bootstrap confidence intervals; because the tail is thin at thirteen episodes and five seeds, the worst episode is reported alongside them. Secondary statistics are raw PnL, excess over HODL, annualised return, Sortino ratio, maximum drawdown, rebalance count, and the cost ledger of fee income, gas, slippage and realised impermanent loss. Per-regime tables stratify by each window's dominant label: bull, bear, sideways or high volatility.

VI.C Ablations and the boundary

The failure modes are ablated one at a time, as set out in Table 8. The cost decomposition is repeated over frictions and on the 0.30 % pool. For the boundary question the runner sweeps realised volatility, volume relative to liquidity, fee tier, gas price and capital, evaluating each point on replayed windows rather than a fitted process.

Decision rules are stated in advance: the learned policy earns a positive verdict only if its mean Sharpe beats the best static width and the hedged HODL with a paired 95 % confidence interval excluding zero,

Table 8: The ablations, each one isolating a design choice the study relies on. Every ablation is trained and evaluated under the protocol of Table 6.

Design choice	Ablation	Question it answers
Band geometry	removed from the observation	does the agent need to see the live band to time rebalances?
Window sampling	windows held fixed instead of resampled	how much does resampling the training windows buy?
Hedge	overlay disabled	what does the hedge contribute net of funding?
Risk penalty	toggled	does the risk penalty change the tail without costing Sharpe?
Regime features	removed	do the regime labels inform the policy?
Frictions	full model, no gas, no slippage, static fee, no impermanent loss	which cost drives the result?
Fee tier	repeated on the 0.30 % pool	does a higher fee tier buy a longer viable range?

and its CVaR is no worse than the best heuristic's. Where the evidence is negative, the negative result is the finding.

VII Results

[PENDING: This section is deliberately a stub while the multi-seed experiments run. Each subsection states the question and the shape of the evidence that will answer it; no number is written until the locked runs exist.]

VII.A Where do fees fail?

The first question is economic. We decompose the passive benchmarks into fee income, impermanent loss, gas and slippage, and locate the conditions under which the losses exceed the fees. [PENDING: Report the decomposition on both fee tiers, the cost of greedy against lazy rebalancing, and where the sign of excess over a passive holding flips across regimes.]

VII.B Why does naive learning fail?

The second question is methodological. A standard policy, trained on the original interface, collapsed to a single action on every seed. We trace that collapse to what the agent could see, what it was rewarded for, and the windows it trained on, and we measure what each repair contributes. [PENDING: Report the pre-fix collapse and the per-fix ablations, with the evidence for each diagnosis.]

VII.C Can one policy do better?

The third question is the central one. [PENDING: Report out-of-sample Sharpe and CVaR for the learned policy against every benchmark, paired per window and across seeds, together with the

action distribution and the decision boundaries it recovers.]

VII.D Where is active management worth it?

The fourth question turns the result into a map. [PENDING: Report the boundary along which active and passive provision achieve equal performance under risk, as a function of volatility, volume relative to liquidity, fee tier, gas and capital, with the analytical LVR threshold overlaid.]

VIII Discussion

[PENDING: The interpretive discussion is deferred until the locked runs land: it will connect the four answers to the active/passive boundary and to the design lessons for learning in this setting. The validity concerns below do not depend on the results.]

The simulator replays recorded events, so prices, liquidity and fee growth are real rather than modelled. Its marginal-agent assumption is the main simplification: the position never moves the price, and its fee share is proportional to active liquidity. That is the standard replay assumption and it is recorded with every run, but it cannot capture queueing at the tick level or the impact of an unusually large position. The gas model charges representative per-action units rather than per-transaction receipts, so gas is a defensible floor rather than an exact quote, and the hedge uses a perpetual proxy whose funding and basis are representative rather than exact for a specific venue.

Table 9 states the threats that bound what the study can claim and the scope each imposes. The parity check bounds the distance between the simulator and the backtester, not the distance between the backtester and reality. Where the data allow, the external part of the gap is quantified by reconciling fees and collects; where they do not, latency, order-flow privacy and MEV are discussed rather than measured. A different design closes the loop by assumption, generating order flow that responds to the fee policy and so identifying counterfactuals the tape cannot provide, at the price of trusting the response model.[18] This report keeps the tape and claims only what the tape supports. We therefore claim results about performance under risk on recorded market states under a stated cost model. We do not claim deployable profitability, optimality of the action grid, or that the policy would survive market conditions absent from 2022 to 2024.

IX Conclusion

Concentrated liquidity promises more fee income while the price stays inside a chosen band, and the same position that earns those fees is the position that loses to informed order flow. An LP who wants to do better than accept that must decide when to recenter, how wide to stay and how much risk to hedge, and each decision is expensive and uncertain.

This report measures whether a risk-sensitive learned policy can make those decisions jointly better than the best rules written by hand, and where active management stops paying. Answering that re-

Table 9: Threats to validity and the scope each imposes.

Threat	Scope
One asset pair at two fee tiers	the boundary is measured within that scope and does not automatically transfer to other pairs
CVaR at 5 % from few episodes	reported with bootstrap intervals and the worst window, not treated as decisive alone
Regime labels from fixed thresholds	a labelling device for reporting and training, not a discovered state
Boundary sweep over several axes	inference rests on paired intervals on the pre-registered primary metric; secondary wins are labelled as such
Simulator and backtester both replay recorded states	claims hold under the stated cost model, not for live execution

quired a world in which costs are measured rather than assumed, so we built a testbed from three years of recorded trades, with an exact replay engine for ground truth, a classical simulator for training, and a parity harness between them. The study then asked four questions: where fees fail, why a standard agent fails, whether a learned policy wins out of sample, and where the boundary between active and passive provision lies.

[PENDING: This paragraph will close with the measured verdicts once the experiments complete. Its shape is fixed: what the four answers were, which of them held, and what the boundary looked like.]

The boundary of the study is also a map for future work: the action space can be extended with continuous width and an explicit exit decision, the boundary sweep can cover more pairs and fee tiers, and latency, private order flow and the exact funding of a real venue can be modelled rather than proxied. Each extension is a re-run of the same testbed rather than a redesign of it, which is the point of building the testbed before asking the question.

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